Income Classification With Naive Bayes

Introduction (51 words)

This report aims to evaluate the performance of a mixed Naive Bayes classifier in predicting the annual income from the 1994 US Census data. I integrated Gaussian and Categorical models to analyse the impact of class imbalance on predictive bias and explore how semi-supervised label propagation influences model confidence and stability.

Question 1 (246 words)

Q1.1 - Class Prior Probabilities

The training data shows a high class imbalance after cleaning. It showed prior probabilities of **P(<=50K) = 0.76 and P(>50K) = 0.24** which is roughly a **3 to 1 ratio in favour of the lower-income class.** Because Naive Bayes factors in the starting probabilities, it will be biased toward the majority group which is reflected in Q2 results where recall for >50K is notably lower (0.5254) than for <=50K (0.9329). The model will need very strong evidence from the features to overcome the low starting prior for high income earners. The model's cautiousness comes at the cost of missing genuinely high-income earners.

Figure 1: Class distribution in the supervised training set.

Q1.2 - Continuous Feature Statistics

Based on the results of the Gaussian density plots, **education-num shows the strongest separation (d = 0.82), followed by age (d = 0.57).** Higher education is often correlated to higher-income as it usually unlocks opportunities for higher-paying specialised roles. Age is also a logical predictor because it represents career experience. Features like **capital-gain** definitely signal wealth, but the values are spread out and will only act as noise for the model.

Feature	Mean (<=50K)	Std (<=50K)	Mean (>50K)	Std (>50K)	Separation (d)
education-num	9.629255	2.447812	11.594551	2.364977	0.816697
age	37.050752	13.711919	43.938495	10.302703	0.573629
hours-per-week	39.428622	11.910803	45.642298	10.396091	0.557108
capital-gain	157.666022	1017.863238	3607.146749	13616.622967	0.471418
capital-loss	55.974580	316.020139	202.362018	603.991020	0.318230

Table 1: Gaussian parameters and class separation.

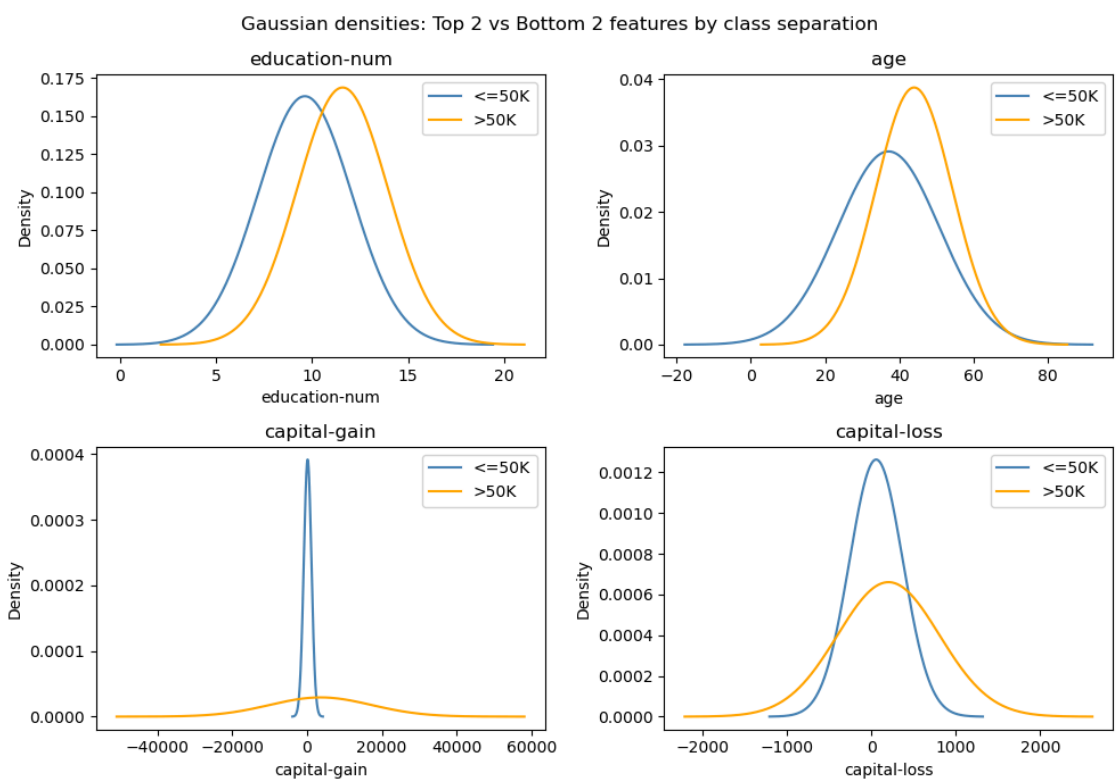

Figure 2: Distribution of Income Classes in the training set.

Q1.3 - Categorical Feature Probability Ratios

The adult dataset shows that education is very highly correlated to higher-income. **Prof-school and Doctorate holders are nearly 8 times more likely to be in the >50K class.** Married-AF-spouse and Taiwan likely have higher R values due to smaller demographics with atypical incomes. Own-child is strongly predictive of a lower income due to children being very expensive to raise and Priv-house-serv is just a low-wage occupation.

Top 5 most predictive of >50K (Highest R)		
Feature	**Value**	**R**
education	Prof-school	7.9578
education	Doctorate	7.1354
marital-status	Married-AF-spouse	4.2882
education	Masters	3.5551
native-country	Taiwan	3.3832

Top 5 most predictive of <=50K (Lowest R)		
Feature	**Value**	**R**
relationship	Own-child	0.0522
occupation	Priv-house-serv	0.0838
occupation	Other-service	0.1354
relationship	Other-relative	0.1445
education	9th	0.1508

Table 2: Top 5 categorical predictors for high income (left) and low income (right).

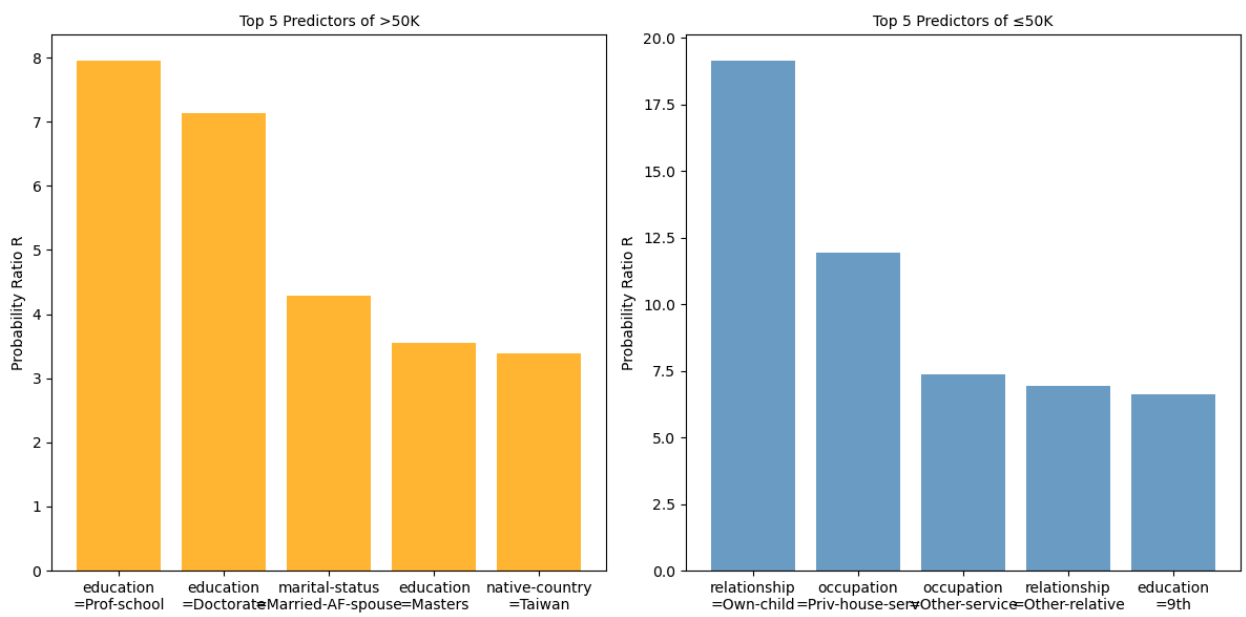

Figure 3: Top 5 categorical predictors for high income (left) and low income (right)

Question 2 (241 words)

Q2.1 - Classification Report

The model has an **83.07% accuracy**. The model is really good at identifying low-income earners (93.3% recall) because they are a majority of the data. However, **it struggles with the high-income group**, catching only 52.54% of them. This is because of the class imbalance noted earlier wherein **the model defaults to the majority class unless the evidence is overwhelming**. The predictions for high earners are usually reliable (72% precision), the low recall is a primary bottleneck.

Class	Precision	Recall	F1-Score	Support
0 (<=50K)	0.8545	0.9329	0.8920	11303
1 (>50K)	0.7240	0.5254	0.6089	3784
Accuracy			0.8307	15087
Macro Avg	0.7892	0.7292	0.7504	15087
Weighted Avg	0.8217	0.8307	0.8210	15087

Table 3: Classification report for the supervised model.

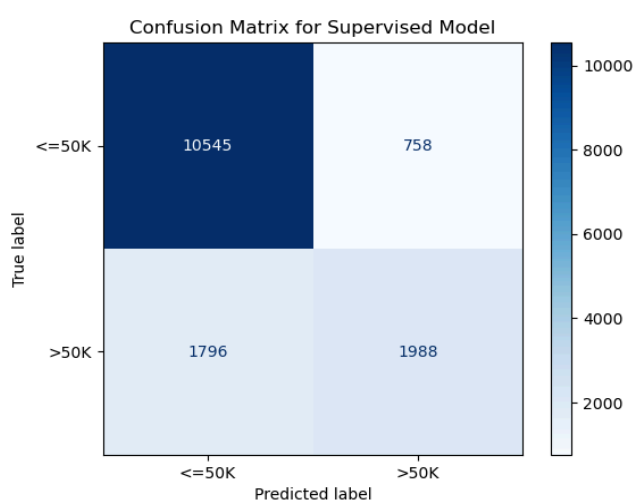

Figure 4: Confusion matrix for the supervised model.

Q2.2 - Out-of-vocabulary Categories

Only one instance in the test set contained an unseen categorical value. This was found within the native-country feature.
I used Laplace smoothing (k = 1) which makes sure that P(xi|cj) remains non-zero for unrecognised attributes. The model classified every row, including the one with the OOV value, by relying on the smoothed probability of known features and overall class priors to generate a valid posterior prediction for each case. No test instances were skipped as the model can still generate a posterior using the class prior and the remaining available feature probabilities.

Q2.3 - Examples of instances classified with high and low confidence.

High Confidence >50K Predictions (Highest R)									
ID	Age	Education	Occupation	Capital-gain	Hours-per-week	Marital-status	True	Pred	R
142	50	Prof-school	Prof-specialty	99999	80	Married-civ-spouse	1	1	1.000000e+300
460	41	Doctorate	Prof-specialty	99999	70	Married-civ-spouse	1	1	1.000000e+300
1050	47	Masters	Exec-managerial	99999	50	Married-civ-spouse	1	1	1.000000e+300

High Confidence <=50K Predictions (smallest R)									
D	Age	Education	Occupation	Capital-gain	Hours-per-week	Marital-status	True	Pred	R
13228	18	10th	Other-service	0	3	Never-married	0	0	9.242193e-10
3216	17	10th	Other-service	0	12	Never-married	0	0	1.074523e-09
10584	17	11th	Other-service	0	5	Never-married	0	0	1.107104e-09

Near Decision Boundary (R~1)									
D	Age	Education	Occupation	Capital-gain	Hours-per-week	Marital-status	True	Pred	R
8144	29	Bachelors	Exec-managerial	0	45	Married-civ-spouse	1	1	1.001370
5292	33	Bachelors	Prof-specialty	0	40	Married-civ-spouse	0	0	0.997216
7037	33	Bachelors	Prof-specialty	0	40	Married-civ-spouse	1	0	0.997216

Table 4: High confidence (top), low confidence (middle), and boundary (bottom) instances.

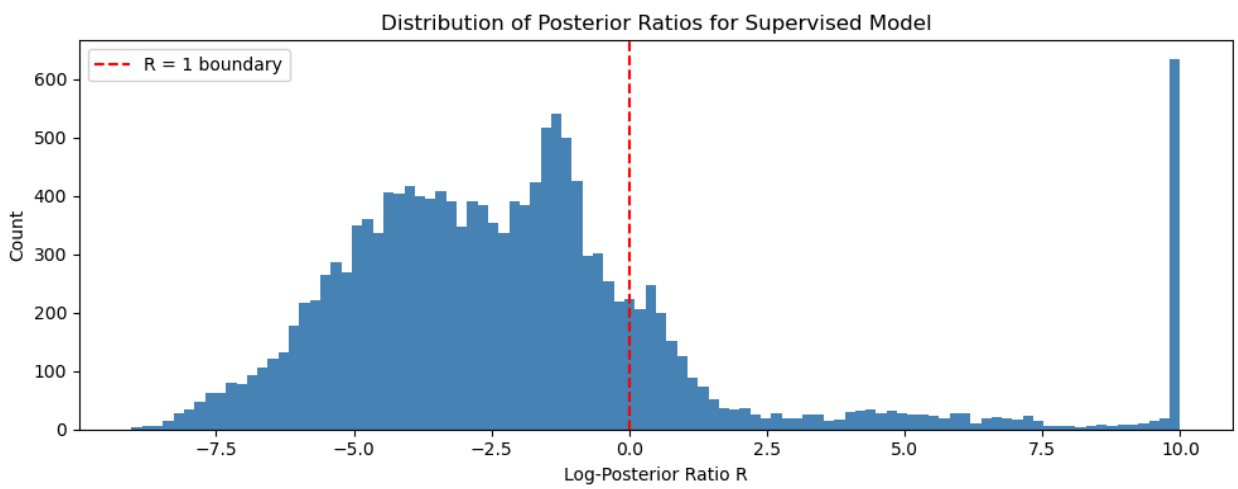

Figure 5: Distribution of posterior ratios around the R = 1 boundary for the supervised model.

THe model shows extreme certainty when features align. The top three showcases are all middle-aged, married, hold advanced degrees, and hold maximal capital gains while the low-income confidence peaks for teenagers working part-time hours. Boundary cases show young, married professionals holding Bachelor’s degrees. Their lack of capital gains makes the model very uncertain and thus leads to misclassifications, particularly for the >50K class where recall is significantly lower.

Question 3 (420 words)

To select the confidence threshold and evaluate the performance of Label propagation, the training set was split into a sub-training set

and a validation set on a 80/20 split. This allowed for tuning without leaking information from the test set.

I chose **Label propagation** over Expectation-Maximisation (EM) or Active Learning because it is fully automated. It also works well with large datasets without needing to manually re-label which simplifies implementation. It is also more straightforward to implement by retraining the combined labelled and pseudo-labelled data. Rather than accepting all pseudo-labels, I only keep instances where posterior ratio R > t or R < 1/t, meaning the model is confident enough in both directions. I also implement an iterative version where the unlabelled data is split 50/50, the first half pseudo-labelled with the base model, the model is retrained, and the update model labels the second half.

Threshold Selection

Threshold	One-Shot (n_pseudo)	One-Shot (val_acc)	Iterative (n_pseudo)	Iterative (val_acc)
1.5	14,632	0.8067	14,668	0.8047
2.0	14,355	0.8067	14,411	0.8050
3.0	13,880	0.8070	14,023	0.8050
5.0	13,288	0.8064	13,530	0.8054
10.0	12,460	0.8060	12,854	0.8057
20.0	11,482	0.8107	11,990	0.8054
50.0	10,332	0.8176	10,737	0.8130
99.0	9,615	0.8206	9,953	0.8173
999.0	7,536	0.8200	7,869	0.8193

Table 5: Validation accuracy vs confidence threshold.

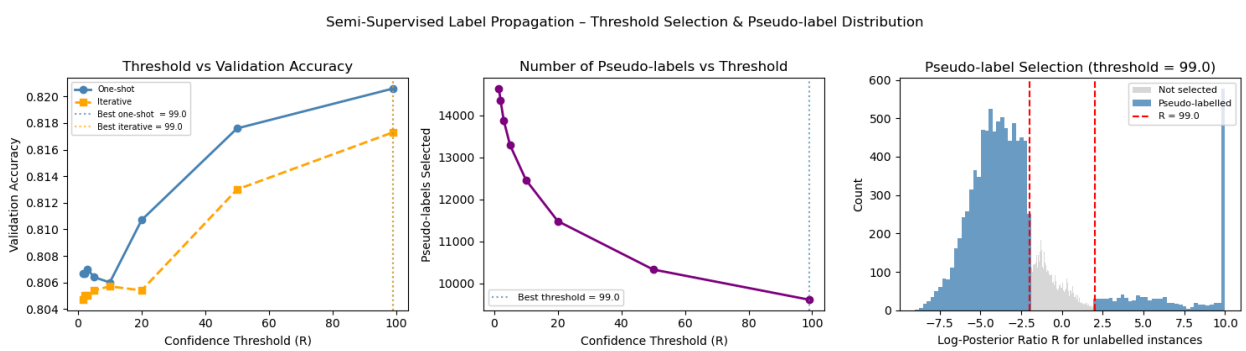

Figure 6: Threshold selection (left), number of pseudo labels for threshold (middle), and pseudo-label selection (right).

The data shows a clear trend wherein the stricter the thresholds (higher R), the higher the validation accuracy. At t = 1.5, around 14,000 pseudo labels were created for both one-shot and iterative but the validation accuracy barely moved (0.8067). At t = 99, only around 9,600 to 10,000 pseudo labels were added but the accuracy is higher at 0.8206 and 0.8173 for one-shot and iterative respectively.

My choice of t = 99 was motivated by the **precision-recall trade-off** where lower thresholds (t < 10) resulted in higher quantities pseudo-labels but likely contained higher levels of noise. This is evidenced by the stagnant validation accuracy of ~80.6%. I prioritised quality over quantity in data in my attempt to mitigate the reinforcement of incorrect class priors.

Figure 6 shows that low-confidence pseudo-labels introduce label noise wherein the less strict the model is, the more pseudo-labels are selected. The model mislabels boundary cases and retrains on them which reinforces those mistakes. High-confidence pseudo-labels are likely to be more correct to a degree. At t = 99, the lower number of pseudo-labels does not have any improvement over the supervised model. The iterative approach performs even worse, possibly due to the fact that the second stage is once again trained on a faulty model, compounding errors.

The final model was trained on the entire 15,076 instances, plus 9,676 pseudo-labelled instances with t = 99. The pseudo-labels were 8,289 for <=50K and 1,387 for >50K. This mirrors and amplifies the original imbalance present in the supervised mode. Therefore, the semi-supervised model trains on an even more lopsided distribution than Q1 did.

Question 4 (471 words)

4.1 Test-Set Performance Comparison

Metric	Q1 (Supervised)	Q3 (Semi-supervised)	Delta
Overall Accuracy	0.8307	0.8190	-0.0117
Precision (>50K)	0.7240	0.6948	-0.0292
Recall (>50K)	0.5254	0.4968	-0.0286
F1-Score (>50K)	0.6089	0.5794	-0.0295
Precision (<=50K)	0.8545	0.8462	-0.0083

Metric	Q1 (Supervised)	Q3 (Semi-supervised)	Delta
Recall (<=50K)	0.9329	0.9269	-0.0060
F1-Score (<=50K)	0.8920	0.8847	-0.0073

Table 6: Supervised vs semi-supervised test-set performance.

The negative delta on every single metric from table 6 shows that the semi-supervised model performs worse than the supervised model. The overall accuracy drops from 83.07% to 81.9% and the F1-Score for >50K drops 2.95%. **This is a clear regression in performance and not simply noise.** The degradation in F1-Score for the high-income class suggests that despite the aggressive filtering of pseudo-labels, they have instead introduced more noise.

As discussed previously, the biggest concern is that label propagation is self-reinforcing mistakes when it is retrained. The supervised model's low recall for the >50K class (0.5254) already shows nearly half of the instances are being mistakenly pseudo-labeled as <=50K. These mislabelled instances only reinforce the model's existing bias. This skewed the training set ratio from roughly 3:1 to 3.28:1, making the imbalance problem worse.

4.2a Supervised vs Semi-Supervised Confidence

The semi-supervised model is paradoxically more confident based on the data. The median posterior shifts from -2.429 to -2.799 for the supervised model and the semi-supervised respectively showing that the median prediction is moving away from the decision boundary in the direction of the low-income earners. The prediction confidence also rises from 83.3% to 87.2%, meaning that the model has become more sure that it has correctly labelled instances when it has actually not. The additional pseudo-labelled data only reinforced the prior toward the majority class, pushing more posteriors further to the left of the R = 1 boundary. **The model is more decisive, but its decisions are more often wrong** for the >50K cases.

4.2b Feature Parameter Changes

Feature	Class	Mean Q1	Mean Q3	Delta Mean	Std Q1	Std Q3	Delta Std
age	<=50K	37.051	36.054	-0.997	13.712	13.767	0.055
age	>50K	43.938	44.089	0.151	10.303	10.562	0.259
education -num	<=50K	9.629	9.466	-0.164	2.448	2.439	-0.009

Feature	Class	Mean Q1	Mean Q3	Delta Mean	Std Q1	Std Q3	Delta Std
education-num	>50K	11.595	11.663	0.069	2.365	2.394	0.029
capital-gain	<=50K	157.666	104.708	-52.958	1017.863	796.413	-221.450
capital-gain	>50K	3607.147	5914.328	2307.181	13616.623	17447.613	3830.990
capital-loss	<=50K	55.975	33.559	-22.416	316.020	245.146	-70.874
capital-loss	>50K	202.362	349.220	146.858	603.991	761.653	157.662
hours-per-week	<=50K	39.429	38.657	-0.771	11.911	11.951	0.040
hours-per-week	>50K	45.642	45.725	0.082	10.396	10.661	0.265

Table 7: Supervised vs semi-supervised feature parameters.

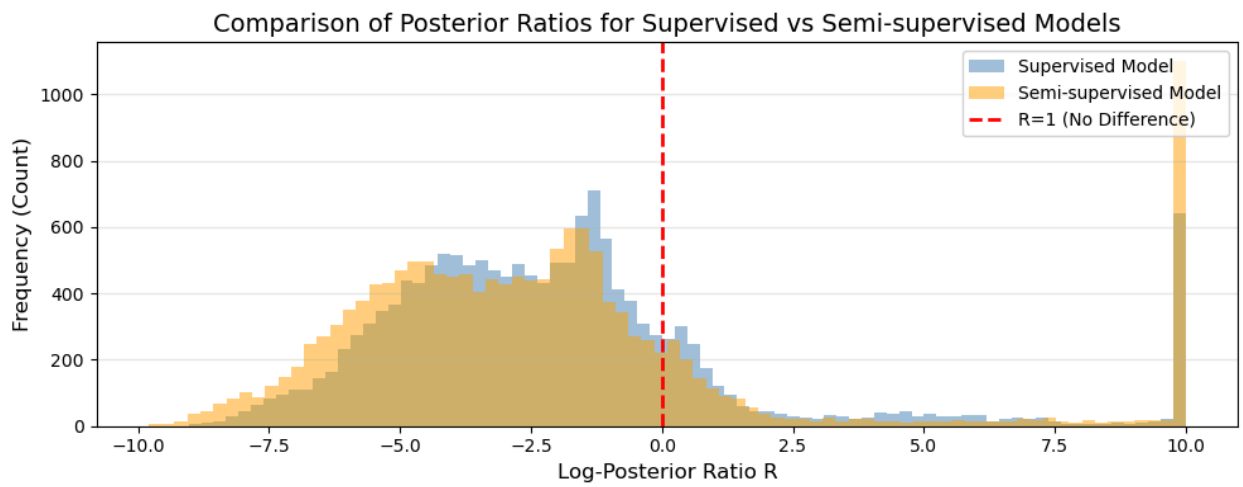

Figure 7: Comparison of supervised and semi-supervised model feature parameters.

For capital gains, the jump in mean and standard deviation for high-income earners and the fall in mean for low-income earners suggests that pseudo labels used as >50K are disproportionately instances with very high capital gains. The model was not confident about them and they thus passed the threshold. Capital-loss also shows a larger shift for >50K than any other feature. Age and education-num have non-noticeable changes, suggesting that pseudo-labelling had little effect on the most reliable features.

4.2c Top Predictors

Top predictors of high income (>50K)

Model	Feature	Value	R
Q1	education	Prof-school	7.9587
Q1	education	Doctorate	7.1354
Q1	marital-status	Married-AF-spouse	4.2882
Q1	education	Masters	3.5551
Q1	native-country	Taiwan	3.3832
Q3	education	Prof-school	13.7503
Q3	education	Doctorate	12.1477
Q3	education	Masters	4.9204
Q3	workclass	Self-emp-inc	4.6491
Q3	native-country	Hungary	3.8368

Top predictors of low income (<=50K)

Model	Feature	Value	R
Q1	relationship	Own-child	0.052
Q1	occupation	Priv-house-serv	0.084
Q1	occupation	Other-service	0.135
Q1	relationship	Other-relative	0.145
Q1	education	9th	0.151

Model	Feature	Value	R
Q3	relationship	Own-child	0.053
Q3	occupation	Priv-house-serv	0.071
Q3	relationship	Other-relative	0.119
Q3	occupation	Other-service	0.123
Q3	native-country	Guatemala	0.128

Table 8: Top 5 predictors of >50K (top) and <=50K (bottom) for both supervised and semi-supercised models.

Table 8 shows that Married-AF-spouse and Taiwan appear in the top 5 and were replaced in the semi-supervised model, by Self-emp-inc and Hungary. Education remains as the top indicator of high-income but the R value differs greatly. Prof-school jumps from 7.69 to 13.75 and Doctorate from 7.14 to 12.15. This once again shows that the pseudo-label the model is most confident in tended to be highly educated professionals with large capital gains.

The <50K predictors were more stable wherein a majority of values are shared between the supervised and the semi-supervised models, though the R values shift slightly. The stability is due to the vast majority of pseudo-labels being <=50K (8,289 out of 9,676). Thus, the model’s understanding of that class is reinforced broadly rather than being reorganised.

Conclusion (57 words)

To conclude, **the semi-supervised label propagation did not improve performance**; the method only reinforced the model’s existing bias. An improvement that can be made to the model would be to investigate class-aware pseudo-labelling and applying different thresholds per class, or exploring other approaches such as Active Learning or Expectation-Maximisation which may be less prone to error compounding.

AI Declaration

Gemini:

- I consulted Gemini to find the theoretical differences between the different semi-supervised models and the pros and cons, along with the syntax for libraries such as sci-kit learn. The logic, implementation, and analysis in this report are my own.
- I used Gemini to assist in debugging errors in numerical stability. This helped me realise the need to use log-space calculations.
- To convert the outputs into markdown to make the tables in my report.

Grammarly:

- I used Grammarly to help me with clarity in the written report.